



Pentagonial

Author: Okba Allaoua

Difficulty: easy

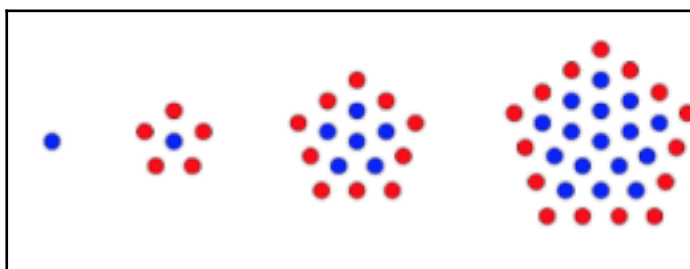
Description

In the ancient lands of **L'INI**, mystics conjure powerful spells by inscribing **Pentagonal Runes**. These magical runes evolve over time, growing stronger with each iteration as more **mana dots** appear in perfect pentagonal harmony.

On the **first level**, only the **core mana dot** exists, pulsing at the center of the rune. As the rune matures, each new level draws a larger pentagonal ring around the previous structure.

- At **level 2**, 6 new dots emerge.
- At **level 3**, the pattern expands to 16 dots.
- By **level 4**, the rune glows with 31 dots.

Deep within the crumbling pages of the **Codex Arcanum**, we discovered a faded image. It shows a mysterious sequence of **Pentagonal Runes**, each surrounded by glowing **mana dots** arranged in a precise, mystical pattern.





Your task, young sorcerer, is to calculate the **total number of mana dots** inscribed in a Pentagonal Rune at the given N th iteration **n**.

Inputs

- Number of iterations that we need to know the number of dots existing.

Outputs

- The number of dots that exist in the whole pentagon on the N th iteration.

Constraints

- If you tried to write the values one by one, trust me, I will know that you've written the values one by one, so don't try it please :)

Examples

Input	Output
3	16
8	141
14	456